



RB003 Week 7 — English Worksheet

Topic: Build Week — Your Full Tycoon · **Course:** Tycoon Engineer · **Time:** about 45 minutes

No new code this week. You connect everything you know: dropper → conveyor → collector → buy button → bigger dropper. You also design your **economy** — the prices and payouts that make your tycoon fun.

1 · Predict 🧠

All three scripts below come from **one tycoon**. Read each one. Before you imagine it running in Studio, write what you think will happen.

```
while true do
  task.wait(2)
  local ore = Instance.new("Part")
  ore.Name = "Ore"
  ore.Size = Vector3.new(1, 1, 1)
  ore.Position = dropper.Position - Vector3.new(0, 3, 0)
  ore.Parent = workspace
end
```

How many ores drop in 10 seconds? What is each one named?


```

local catalog = {
    {name = "Ore", payout = 5},
    {name = "Gold", payout = 20},
}

collector.Touched:Connect(function(hit)
    for _, item in ipairs(catalog) do
        if hit.Name == item.name then
            cash.Value += item.payout
            hit:Destroy()
        end
    end
end)

```

A part named "Gold" touches the collector. How much cash does the player get? What happens to a part named "Rock"?


```

local buying = false

pad.Touched:Connect(function(hit)
    if buying then return end
    buying = true

    if cash.Value >= 100 then
        cash.Value -= 100
        local model = ServerStorage.GoldDropper:Clone()
        model.Parent = workspace
    end

    task.wait(1)
    buying = false
end)

```

Each Ore pays 5. How many ores must a player collect before they can buy this upgrade? After buying, how much cash is left?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The player just bought the GoldDropper. The collector should collect **Ore AND Gold**. Right now gold parts pile up on the conveyor and cash never goes up for them.

```
collector.Touched:Connect(function(hit)
    if hit.Name == "Ore" then
        cash.Value += 5
        hit:Destroy()
    end
end)
```

Hint: use the catalog with `ipairs`, like in Predict.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The pad's sign says "**Gold Dropper — 100**". Buying should unlock the GoldDropper. Right now players pay 100 and get a second ore dropper instead.

```
if cash.Value >= 100 then
    cash.Value -= 100
    local model = ServerStorage.OreDropper:Clone()
    model.Parent = workspace
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The dropper and the collector should use **the same name**. Right now the collector never pays, because the names do not match.

```
-- dropper script
local ore = Instance.new("Part")
ore.Name = "ore"

-- collector script
if hit.Name == "Ore" then
    cash.Value += 5
    hit:Destroy()
end
```

Hint: names must match exactly — capital letters count.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your tycoon in Roblox Studio. First **design your economy**, then build the full chain: dropper, conveyor, collector, and buy buttons that unlock upgrades in order.

Plan your upgrades before you build. Fill in this table:

Upgrade	Price	What it does
1.		
2.		
3.		

Check your balance: if your first drop pays 5 and your first upgrade costs 100, a player needs 20 drops. Do the same math for **your** numbers — how many drops until a player can buy your first upgrade? Is that fast enough to be fun?

When your tycoon works, send a photo or video of it running, then explain what you built. Use these sentence starters — write 4 to 6 sentences total.

My money loop works like this: ...

My first upgrade costs ... because ...

Players buy my upgrades in this order: ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

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4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you **playtest** your tycoon from the start. Talk through it like you are teaching someone who has never seen it. Try to use these words: **money loop, payout, price, upgrade, balance**.

1. Start with 0 cash. Collect drops and show the cash going up.
2. Buy your first upgrade and show what it unlocks.
3. Show that the new parts get collected too — nothing piles up.
4. Say one price or payout you changed to make the game more fun, and why.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.